Chapter 1: Linear Systems

Worked Answer Key Part 1 of 5

Solutions correspond to Exercises 1–20 in the companion set. All calculations use exact arithmetic.

A. Row, column, and matrix pictures

1. $A = \begin{pmatrix} 1 & -2 \\ 3 & 1 \end{pmatrix}$, $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, and $\mathbf{b} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$. The column equation is

$$x \begin{pmatrix} 1 \\ 3 \end{pmatrix} + y \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

From $x = 4 + 2y$, substitution gives $12 + 7y = 5$. Thus $y = -1$ and $x = 2$. The lines $y = x/2 - 2$ and $y = 5 - 3x$ intersect at $(2, -1)$.

2. The row products are $2 - 2 - 3 = -3$, $-3 + 12 = 9$, and $4 + 1 + 3 = 8$. The column calculation gives

$$2 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} + 3 \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 9 \\ 8 \end{pmatrix}.$$

Both methods implement the definition of matrix-vector multiplication.

3. (a) The two equations describe the same line. All solutions are $(x, y) = (2 - t, t)$, $t \in \mathbb{R}$. The right-hand side $(2, 4)^T$ is twice either column $(1, 2)^T$.

(b) The equations describe distinct parallel lines, so there is no solution. Every combination of the columns has the form $(c, 2c)^T$, whereas $(2, 5)^T$ does not.

4. Subtracting the second equation from the first gives $2y = 2$, so $y = 1$. Then $x + z = 2$ and $2x + z = 4$, giving $x = 2$, $z = 0$. The planes have the unique common point $(2, 1, 0)$. Verification:

$$2 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}.$$

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B. Gaussian elimination and zero pivots

5. Start with $[A \mid \mathbf{b}]$. Apply $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 + R_1$, and then $R_3 \leftarrow R_3 - 2R_2$:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 2 \\ 0 & 3 & -1 & 4 \\ 0 & 6 & -3 & 3 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 2 & 1 & -1 & 2 \\ 0 & 3 & -1 & 4 \\ 0 & 0 & -1 & -5 \end{array}\right).$$

   The pivots are $2, 3, -1$. Back substitution gives $z = 5$, $3y - z = 4$, and $2x + y - z = 2$, hence $(x, y, z) = (2, 3, 5)$.

6. After $R_1 \leftrightarrow R_2$, use $R_3 \leftarrow R_3 - 2R_1$ and $R_3 \leftarrow R_3 - \frac{1}{2}R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 7/2 & 15/2 \end{array}\right).$$

   Thus $z = 15/7$, $y = 10/7$, and $x = 11/7$. All three pivots are nonzero after the exchange. A zero in one pivot position may simply require reordering the equations.

7. The operations $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 + R_1$ produce rows $[0\ 0\ 0 \mid 0]$ and $[0\ 0\ 0 \mid 4]$. The latter says $0 = 4$, so the original system is inconsistent.

   With the last right-hand side equal to -1, both lower rows are zero. Set $y = s$, $z = t$; then all solutions are $(1 - 2s + t, s, t)$, with $s, t \in \mathbb{R}$.

8. Subtract twice the first equation from the second to obtain $(a - 4)y = b - 6$.

   - If $a \neq 4$, the unique solution is $y = (b - 6)/(a - 4)$, $x = 3 - 2(b - 6)/(a - 4)$.
   - If $a = 4$ and $b = 6$, all solutions are $(x, y) = (3 - 2t, t)$.
   - If $a = 4$ and $b \neq 6$, there is no solution.

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C. Elementary and permutation matrices

9.
$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix}, \qquad E^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{pmatrix}.$$

Multiplication gives $E^{-1}E = I$. The rows of EA are respectively row 1 of A, row 2 of A, and row 3 minus four times row 1 of A.

10.
$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}.$$
$$E_3E_2E_1A = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & -1 \end{pmatrix} = U.$$

The rightmost factor acts first. In particular, E_3 uses the already updated second row. Matrix multiplication is generally not commutative, so reversing the factors need not give U.

11.
$$PB = \begin{pmatrix} 7 & 8 & 9 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \qquad P^{-1} = P^T = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

The new row order is $(3, 1, 2)$. The inverse restores the original order, and direct multiplication gives $P^{-1}P = I$.

12. The columns of P are the standard basis vectors in some order. Their dot products are 1 for identical columns and 0 for distinct columns, so $P^TP = I$. Thus $P^{-1} = P^T$.

The matrix in Exercise 11 is an example: its square is $P^T \neq I$, while $P^3 = I$. A cycle of length three requires the inverse cycle to undo it; repeating it twice in total does not restore the original ordering.

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D. LU factorization and pivoting

13. The elimination multipliers are $2, -1, 2$, so

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & -1 \end{pmatrix}.$$

The rows of LU are $(2, 1, -1)$, $(4, 5, -3)$, and $(-2, 5, -2)$, as required. Solve $L\mathbf{y} = \mathbf{b}$: $y_1 = 2$, $y_2 = 8 - 4 = 4$, $y_3 = 1 + 2 - 8 = -5$. Then $U\mathbf{x} = (2, 4, -5)^T$ gives $(x, y, z) = (2, 3, 5)$.

14.

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 1/2 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 7/2 \end{pmatrix}.$$

After the exchange, subtract twice row 1 from row 3, then half of row 2 from row 3. Direct multiplication gives

$$LU = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 4 \end{pmatrix} = PB.$$

15. Without pivoting, the multiplier is $1/0.01 = 100$. Partial pivoting exchanges the rows, giving

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ 0.01 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 1 \\ 0 & 0.99 \end{pmatrix}.$$

The multiplier is now 0.01. Since $P\mathbf{b} = (2, 1.01)^T$, forward substitution gives $(2, 0.99)^T$, and back substitution gives $\mathbf{x} = (1, 1)^T$. Smaller multipliers help limit amplification of rounding errors during elimination; exact arithmetic gives the same solution either way.

16. Forward substitution in $L\mathbf{y} = \mathbf{c}$ gives

$$y_1 = 3, \quad y_2 = 7 - 2(3) = 1, \quad y_3 = 0 + 3 - 2(1) = 1.$$

Back substitution in $U\mathbf{w} = \mathbf{y}$ gives $w_3 = -1$, $3w_2 - w_3 = 1$, and $2w_1 + w_2 - w_3 = 3$. Thus $\mathbf{w} = (1, 0, -1)^T$. The factors are computed once; each additional right-hand side requires only two triangular solves.

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E. Inverses and computational cost

17. Starting with $[D \mid I]$, use $R_1 \leftarrow R_1/2$, $R_2 \leftarrow R_2 - 3R_1$, $R_2 \leftarrow 2R_2$, and $R_1 \leftarrow R_1 - \frac{1}{2}R_2$:

$$\left(\begin{array}{cc|cc} 1 & 1/2 & 1/2 & 0 \\ 0 & 1/2 & -3/2 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{array}\right).$$

Thus $D^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. Multiplication gives $DD^{-1} = I$, and $\mathbf{x} = D^{-1}(1,0)^T = (2,-3)^T$.

18. Use $R_3 \leftarrow R_3 - R_1$ and then $R_3 \leftarrow R_3 + 2R_2$:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & -1 & 2 & 1 \end{array}\right).$$

Next use $R_3 \leftarrow R_3/3$, $R_2 \leftarrow R_2 - R_3$, and $R_1 \leftarrow R_1 - 2R_2$. This gives $[I \mid C^{-1}]$, where

$$C^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{pmatrix}.$$

Multiplying C by the displayed integer matrix gives $3I$, verifying the inverse.

19. Associativity gives

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = I, \qquad (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = I.$$

For the example, $AB = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ and

$$B^{-1}A^{-1} = \begin{pmatrix} 1/2 & -1/2 \\ 0 & 1 \end{pmatrix}, \qquad A^{-1}B^{-1} = \begin{pmatrix} 1/2 & -1 \\ 0 & 1 \end{pmatrix}.$$

Multiplying AB by the second candidate gives $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \neq I$. The order of the inverse factors matters.

20. (a) Reusing the factors costs approximately $\frac{2}{3}n^3 + 2kn^2$.

(b) Repeating factorization costs approximately $k(\frac{2}{3}n^3 + 2n^2)$.

(c) At $n = 100$, $k = 10$, these estimates are about 866,667 and 6,866,667 operations, respectively. Repeating factorization costs about 7.92 times as much.

(d) The leading factorization cost increases by a factor of $2^3 = 8$. A tridiagonal matrix has nonzero entries only on the main diagonal and the two adjacent diagonals. A suitable elimination method can exploit this structure and avoid most operations required for a dense matrix.